Neil Janwani

Doctoral Student, Georgia Institute of Technology, Atlanta, Georgia, USA
njanwani@gatech.edu — (989) 423-4188 — njanwani.github.io — www.linkedin.com/in/neil-janwani-1b71a4129/

EDUCATION

PH.D. **Robotics**, Georgia Institute of Technology — AUG 2024 — JUN 2029
Principal Advisor: Prof. Maegan Tucker (Dynamic Mobility Lab). — GPA: 4.0
NSF Graduate Research Fellowship, Georgia Tech IRIM Fellowship

B.S. **Computer Science**, California Institute of Technology — SEP 2020 — JUN 2024
Research Advisors: Prof. Aaron Ames, Prof. Joel Burdick. — GPA: 4.0

RESEARCH SPOTLIGHT

Multi-Objective Robotic Optimization. JUN 2025 — Present
Generating high-dimensional Pareto frontiers enables diverse personalization and tuning of learned controllers post-training.

- Generated diverse, Pareto-optimal whole-body locomotion across six objectives in ∼2 hours for the BRUCE humanoid.
- Leveraged hypernetworks to tractably approximate high-dimensional Pareto sets using multi-objective PPO.
- Created GPU-native Multi-Objective RL Python toolbox achieving up to 270× speedups over prior work [1].

Robust and Stylized Humanoid Locomotion. JAN 2025 — Present
Encoding when and how humanoid controllers should deviate from idealized reference motions improves robustness.

- Improved imitation accuracy and adherence to gait style via an RL policy that selects which reference to track [2].
- Built a continuous gait library by smoothly interpolating between dynamically feasible reference motion pairs [2].
- Developing an adaptive framework for varying the trade-off between imitation-fidelity and robustness for humanoids.
- Achieved hardware deployment of aforementioned algorithms on the G1 Humanoid and BRUCE humanoid robots.

Human-Robot Interaction and Personalization. JUN 2021 — Present
Robotic personalization and aligned human-robot teams can improve robot integration in the real world.

- Devised quality-diverse exploration stage to familiarize users with their options before eliciting their preferences.
- Showed that initial exploration increases preference alignment on a synthetic oracle with non-stationary preferences.
- Improved operator-robot alignment without losing formal safety guarantees with an intention-aware CBF [3].
- Designed motor control and gait-phase inference of a soft ankle exoskeleton and ran human-subject experiments [4].

PROFESSIONAL EXPERIENCE

Burdick Group, Caltech — Pasadena, California
Research Fellow: Kiyo and Eiko Tomiyasu Named Scholar — JUN 2023 — JUN 2024

- Devised CBF-based safety filters that preserved formal safety guarantees while allowing operator-specific personalization.
- Developed a haptic GUI communicating live robot safety data for Caltech's LINC DARPA team.
- Presented work at ICRA 2024 in Yokohama, Japan [3].

MIT Lincoln Laboratory — Lexington, Massachusetts
Research Intern — JUN 2022 — SEP 2022

- Formulated additions to linear time-invariant control methodology for sensorless control of gimbal motor.
- Refactored Simulink and Jupyter API for experimentation of sensorless control algorithms on gimbal testbed.
- Achieved accurate sensorless estimation of steady-state electrical phase in under 20 seconds.

AMBER Lab, Caltech — Pasadena, California
Research Fellow — JUN 2021 — JUN 2024

- Learned LSTM gait-phase approximator from data collected via ankle-mounted IMUs for stride-based control.
- Designed and built a custom ankle exoskeleton, including PCB design, biometric sensor integration, FOC-based motor control, and safety-critical hardware constraints for wearable use [4].

TECHNICAL SKILLS

- **Programming Languages:** `Python, C/C++`
- **Software:** JAX, PyTorch, OpenCV, CVXPY, MuJoCo, Gymnasium, MjLab, Isaac Lab, ROS1&2, (Ignition) Gazebo, Linux, Git, CMake, MATLAB/Simulink, Mathematica, Solidworks, Weights and Biases, SimpleFOC
- **Hardware:** Unitree G1 Humanoid, Westwood Robotics BRUCE Humanoid, Wandercraft Atalante Exoskeleton, (Custom) Ankle Exoskeleton, DEVCOM GVR-BOT, (Custom) 6-DOF Robotic Manipulator

PUBLICATIONS

Journal Publications:

[1] **Janwani N.**, Novoseller E., Lawhern V., Tucker M. (2026). "MO-Playground: Massively Parallelized Multi-Objective Reinforcement Learning for Robotics." *IEEE Robotics and Automation Letters, 2026* [Webpage]

Conference Publications:

[2] **Janwani N.**†, Madabushi V†., Tucker M. "NaviGait: Navigating Dynamically Feasible Gait Libraries using Deep Reinforcement Learning." *IEEE International Conference on Robotics and Automation (ICRA), 2026* [Webpage]

[3] **Janwani, N.**†, Daş, E†., Touma, T., Wei, S. X., Molnar, T. G., & Burdick, J. W. "A Learning-Based Framework for Safe Human-Robot Collaboration with Multiple Backup Control Barrier Functions." *IEEE International Conference on Robotics and Automation (ICRA), 2024* [Paper] [Video]

Patents:

[4] Shaikewitz L., Tucker M., **Janwani N.**, Ames A., "Front-leg assistive exoskeleton" *US20240245569A1* [Paper]

TEACHING EXPERIENCE

Single Board Computers in Research, Caltech CS 12 Course Pasadena, California
Course founder and instructor JAN 2023 — MAR 2024

- Developed 10-week course on prototyping robotic and computing systems for research applications.
- Taught 50 students in concurrent software and circuit design, sensor interfacing, microcontroller communication.
- Wrote hands-on labs in basic circuitry, hardware input, motor control and coached groups through final projects.
- Hired and trained an undergraduate teaching assistant to assist with course activities and grading.
- Granted $3000 from Caltech in funding for larger iteration of the course for the 2024 cycle

Caltech Robotics and Computer Science Pasadena, California
Teaching Assistant JAN 2022 — JUN 2024

- Designed a mobile robot as Robotics Head Teaching Assistant for a 35 student diversity, equity and inclusion program.
- Held office hours and wrote exam materials for classes of 100+ students in Experimental Robotics (ROS2: Python), Kinematics (ROS2: Python), Data Structures (Java), Software Design (C), and Computing Systems (x86 Assembly).

AWARDS

NSF Graduate Research Fellowship APR 2024
Fellowship awarded by the National Science Foundation (NSF). Atlanta, Georgia

IRIM Graduate Research Fellowship APR 2024
Fellowship awarded by Georgia Tech's Institute of Robotics and Intelligent Machines (IRIM). Atlanta, Georgia

Thomas A. Tisch Prize for Undergraduate Teaching in Computing and Mathematical Sciences APR 2024
Awarded for development and instruction of new robotics course at Caltech. Pasadena, California

Kiyo and Eiko Tomiyasu Research Fellowship JUL 2023
Awarded a fully funded summer undergraduate research at the Burdick Group at Caltech. Pasadena, California

Housner Fund DEC 2023
Granted $3300 to develop a Caltech robotics course, supporting increased student interest. Pasadena, California

SELECTED PROJECTS

- **Socially-Reasonable Humanoid Gestures:** Collected dataset of human social gestures and retargeted to Unitree G1 humanoid robot. Utilized an LLM as a high-level social task planner to sequence retargeted motions.
- **Learning Humanoid Motion Priors Online:** Leveraged VLMs to generate high-level task plans from natural language goals, and efficiently request one-shot demonstrations when the robot's skill library is insufficient.
- **Human-Robot Manipulation:** 5-DOF manipulator to play *Carrom* in a dynamic environment with human opponents.
- **Wikipedia Agent**: Sub-agents for retrieval augmented generation. Improved answer accuracy and speed over baselines.
- **AI Diagnostician:** Trained large vision models to predict pathologies from "CheXpert" chest X-rays.
- **Exploration-Based Planner:** Information-based exploration of an unknown space with probabilistic sensors.
- **Graph Search for Replanning:** Compared efficiency of D* from A* and Dijkstra for replanning on large graphs.
- **Quadruped Kinematics:** Wrote forward/inverse kinematics to simulate a quadrupedal robot designed in Solidworks.
- **Hierarchical Quadruped Planner:** Wrote hierarchical A* planner for quadrupedal locomotion on diverse terrains.
- **Mobile Robotics:** Implemented localization, obstacle avoidance, and odometry algorithms on hardware.